

UTKARSH ANAND

M.Sc. Robotic Systems Engineering
Autonomous & Legged Robotic Systems

✉ utkarsh.anand@rwth-aachen.de
✉ utkarshanand221@gmail.com
in /utkarsh-anand
@utkarshanand140
🌐 Personal Website



About Me

Robotics graduate student with industry experience in quadruped robotic systems and embedded control architectures. Interested in robust autonomous systems operating in unstructured and extreme environments, with a focus on system-level integration of control, estimation, and energy-aware design.

Education

RWTH Aachen University

M.Sc. Robotics Systems Engineering

Coursework: Robotic Systems, Robot Kinematics & Dynamics, Machine Learning, Control Systems

2025 - PRESENT

AACHEN, GERMANY

Manipal Institute of Technology

B.Tech. Electrical & Electronics Engineering

2020 - 2024

KARNATAKA, INDIA

Experience

Strider Robotics, ARTPARK (IISc)

Associate Electrical Systems Engineer

JULY 2024 – JULY 2025

BENGALURU, INDIA

- ▶ Contributed to system-level electrical and embedded architecture of heavy-duty quadruped robots for real world employment.
- ▶ Designed and integrated battery management systems for high transient locomotion loads.
- ▶ Developed RTOS-based firmware for distributed robotic subsystems.
- ▶ Implemented safety-critical RF-based emergency stop systems.
- ▶ Collaborated across hardware and locomotion teams to ensure robust field performance.

Robotics Embedded Systems Intern

JAN 2024 – JUN 2024

- ▶ Designed next-generation battery systems for quadruped robots ([Bachelor Thesis](#)).
- ▶ Integrated CAN/CAN-FD communication for real-time distributed control.
- ▶ Developed embedded firmware for robotic power and control subsystems.

Boltzmann Labs

NOV 2023 – DEC 2023

Deep Learning Research Intern

REMOTE

- ▶ Developed reward-based LLMs for the generation of peptides with a high protein-protein interaction binding affinity.
- ▶ Created prediction models for T-Cell Receptor-Peptide binding & Peptide-HLA Class-I binding for high Spearman Correlation Scores.

thrustMIT Student Rocketry Team

OCT 2021 – SEPT 2023

Payload Head & Launch Operations Lead

MANIPAL, INDIA

- ▶ Led development of a 3-DOF parallel manipulator (STEWIE) for stabilizing a CubeSat payload during sounding rocket flight.

- ▶ Designed and implemented closed-loop PID control with Kalman filter-based state estimation under real-time constraints.
- ▶ Conducted hardware–software co-design and validation under vibration-intensive launch conditions.
- ▶ Oversaw system integration and launch operations at Spaceport America Cup (USA).

Patents

SMART SCRUB FOR POSTURE CORRECTION

AUG 2023

Indian Patent Office — Published

APP. NO. 202341056831

- ▶ Smart wearable scrubs for pain relief & posture correction for medical professionals.
- ▶ Developed complete technology for the product including hardware and firmware.

PAYLOAD DEPLOYMENT MECHANISM FOR SOUNDING ROCKETS

NOV 2023

Indian Patent Office — Published

APP. NO. 202341075776

- ▶ Novel radial deployment mechanism for payloads in sounding rockets without pyrotechnics.
- ▶ Contributed in complete design, development & testing for mechanism.

Projects

STEWIE: A parallel manipulator robot

- ▶ Developed 3-DOF stabilization mechanism for CubeSat payload in sounding rocket launch.
- ▶ Designed closed-loop PID controller with Kalman filter-based state estimation.
- ▶ Conducted hardware–software co-design and validation under vibration-intensive conditions.

VisionTag: Auto-Annotation Software

- ▶ Developed Python + OpenCV-based annotation tool to accelerate dataset preparation.
- ▶ Reduced manual annotation time by 96% for confidential defense datasets.

Skills

Programming Python, C/C++/Embedded C, MATLAB

Robotics State Estimation, Control Systems, RTOS, CAN/CAN-FD

Hardware STM32, PCB Design(KiCAD)

Language English (Professional Fluency)

Research Interests

- ▶ Robust autonomous robotic systems
- ▶ System-level integration of control and estimation
- ▶ Legged robotics in unstructured environments
- ▶ Space robotics and extreme-environment autonomy
- ▶ Energy-aware and safety-critical robotic architectures